

Chapter 29 Back Matter Check

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29 Beyond Perturbation Theory

Problems

1. What is the Witten index of the model in Problem 4 of Chapter 26 for $a \neq 0$? For $a = 0$? Can supersymmetry be broken by higher-order effects in this model for $a = 0$? Explain.
2. Consider the renormalizable supersymmetric theory with $SO(N_c)$ gauge symmetry and N_f left-chiral scalar superfields Φ_n in the N -vector representation. What can you say about the structure of the non-perturbative Wilsonian Lagrangian density when the bare superpotential vanishes? What if there is a bare superpotential $\sum_n \Phi_n \Phi_n$?
3. In the theory with Lagrangian density (29.5.22), what is the relation between the component fields of the spinor field strength superfield W_α and its dual $\widetilde{W}_\alpha$?

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